Review of Transmission Lines – Part 1

James T. Aberle, Ph.D.
Associate Professor Emeritus

School of ECEE

Ira A. Fulton Schools of Engineering
Arizona State University

Lecture Objective

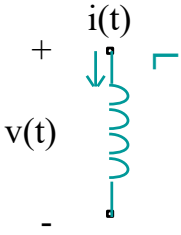
- By the end of this lecture, you will :
  - Recall important concepts for analyzing transmission lines including the telegrapher's equations, wave equations, and transmission line parameters.

Overview of Transmission Lines

- Physically, a ***transmission line*** is a device which guides the transmission of electromagnetic energy from one point to another in a TEM or quasi-TEM mode.
- The ideal transmission line model is a *distributed* circuit element wherein the voltages and currents are functions of position as well as time.
- The voltages and currents within the ideal transmission line circuit element are solutions to wave equations.
- The ideal transmission line model can be used to model physical transmission lines as well as provide a model for the higher order modes (TE, TM) of a waveguide.

Lumped Circuit Element Models

- Lumped circuit elements are defined by *ordinary*, *linear*, *constant-coefficient* differential equations.
- The differential equations obtained for the circuit using KVL and/or KCL (i.e., loop or nodal analysis) involve only one variable: time.
- We are really assuming that the entire circuit is at a single point in space, or equivalently, that the travel time of the electric signal is negligible.



$$v(t) = L\frac{di(t)}{dt}$$

Distributed Circuit Elements

- A ***distributed circuit*** is one for which the travel time of the electric signal between the components cannot be neglected.
- The voltages and currents in such a circuit are functions of *position* as well as *time*.

Lumped Versus Distributed Circuit Element

Points of view for determining whether a circuit should be considered “lumped” or “distributed”:

- Rise time versus travel time
- Period versus travel time
- Component size versus wavelength

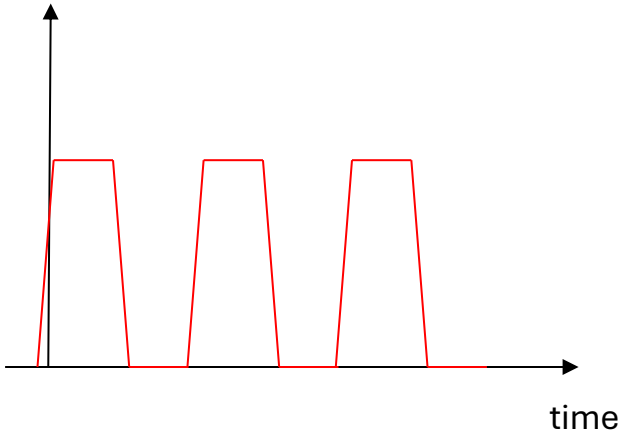
Rise Time Versus Travel Time

- ***Rise time*** **(***RT***)** is the time required for the signal to change from some low percentage to some high percentage of its final value (often 10% to 90% or 20% to 80%).
- ***Travel time*** or ***delay time*** **(***TD***)** is the one-way propagation delay through the signal path.
- $TD = l/v_p$ where l is the length of the signal path and v_p is the velocity of propagation.
- Rule of thumb:
  - $RT/TD > 10 \Rightarrow$ ***definitely lumped***
  - $RT/TD < 1 \Rightarrow$ ***definitely distributed***

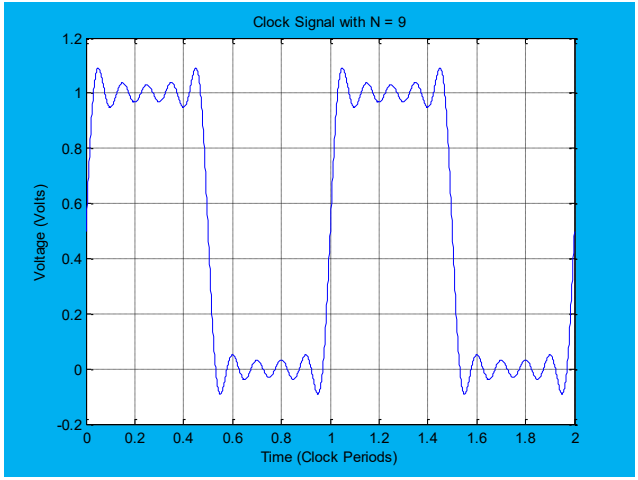
This rule-of-thumb is useful for signal integrity analysis of digital signals.

Question

- What are some reasons for considering 10% to 90% (or 20% to 80%) of the rise time rather than 0% to 100%?



Ideal clock signal



More realistic clock signal

Period Versus Travel Time

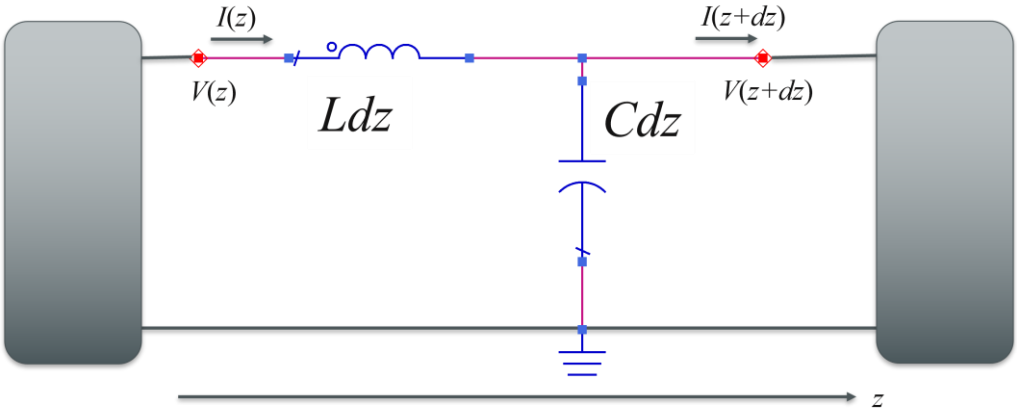
- The ***period*** **(*T*)** of a sinusoidal signal is given by *T=1/f* where *f* is its frequency; the ***period*** (***T***) of a digital signal can be viewed as reciprocal of the symbol rate.
- Rule-of-thumb:
  - $TD/T < 0.01 \Rightarrow$ ***definitely lumped***
  - $TD/T > 0.1 \Rightarrow$ ***definitely distributed***

Component Size Versus Wavelength

- Rule of thumb:
  - $l/\lambda < 0.01 \Rightarrow$ ***definitely lumped***
  - $l/\lambda > 0.1 \Rightarrow$ ***definitely distributed***
- where l is the largest dimension of the component and $\lambda = c/f$ is the wavelength

This rule-of-thumb is only useful for narrow band pass signals with a well-defined carrier frequency whose wavelength can be used in its application.

Equivalent Circuit Representation of the Telegrapher's Equations (1 of 4)



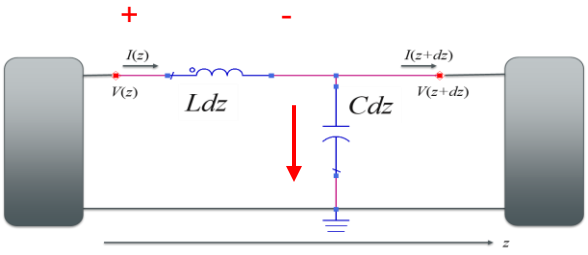
Equivalent Circuit Representation of the Telegrapher's Equations (2 of 4)

- The volt drop across the inductor in the direction of assumed current flow $I(z)$ is:

$$V(z) - V(z + dz) = j\omega L dz I(z)$$

- The current through the capacitor from the top node to ground is:

$$I(z) - I(z + dz) = j\omega C dz V(z + dz)$$



Equivalent Circuit Representation of the Telegrapher's Equations (3 of 4)

$$\frac{V(z+dz)-V(z)}{dz} = -j\omega L I(z)$$

$$\frac{I(z+dz)-I(z)}{dz} = -j\omega C V(z+dz)$$

Equivalent Circuit Representation of the Telegrapher's Equations (4 of 4)

- Take the limit as $dz \rightarrow 0$:

$$\boxed{\frac{dV}{dz} = -j\omega LI}$$

$$\boxed{\frac{dI}{dz} = -j\omega CV}$$

The telegrapher's equations apply to any distributed circuit which can be characterized in terms of a series inductance per unit length and a shunt capacitance per unit length.

Wave Equations and Solutions - From Telegrapher's Equations

$$\frac{d^2V}{dz^2} = -j\omega L\frac{dI}{dz} = -\omega^2 LCV \quad \Rightarrow \quad \frac{d^2V}{dz^2} + \beta^2 V = 0$$

$$\boxed{V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{+j\beta z}}$$

$$\beta = \omega\sqrt{LC}$$

$$I(z) = \frac{-1}{j\omega L}\frac{dV(z)}{dz} = \frac{\beta}{\omega L}V_0^+ e^{-j\beta z} - \frac{\beta}{\omega L}V_0^- e^{j\beta z}$$

$$\boxed{I(z) = \frac{V_0^+}{Z_0}e^{-j\beta z} - \frac{V_0^-}{Z_0}e^{j\beta z}}$$

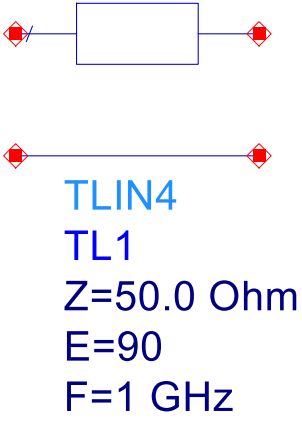
$$Z_0 = \sqrt{\frac{L}{C}}$$

Implications of Above

- The telegrapher's equations hold for any distributed circuit that can be characterized by a series inductance per unit length and a shunt capacitance per unit length.
- As we shall see, the telegrapher's equations can be generalized to distributed circuits that can be characterized by a series impedance per unit length and a shunt admittance per unit length.

(Lossless)Transmission Line Circuit Element

- Transmission line circuit element is specified by
  - Characteristic impedance (Z in the ADS schematic)
  - Electrical length βL (E in the ADS schematic expressed in degrees)
  - Frequency at which electrical length is specified (F in the ADS schematic).



Schematic symbol for transmission line circuit element in Agilent ADS software.

Summary

- Transmission lines are distributed circuit elements: voltage and current are functions not only of time but of position as well.
- Telegrapher's equations are obtained by considering an infinitesimal length of transmission line.
- Solutions to the telegrapher's equations are waves propagating on the transmission line.

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